

BASED ON THE PAPER

Grant Selection Design and Start-Up Innovation

Evidence from Design-Dependent Subsidy Effects

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Curriculum Vitae

▶ School

Sidcot Independent School England	2012–2014
International Baccalaureate	

▶ University

Technical University of Munich B.Sc. Technology and Management	2014–2020
Technical University of Munich M.Sc. Technology and Management	2020–2024

▶ Professional Experience

ColInvest Corporate Finance Analyst	2021–2024
agentix GmbH Co-Founder & COO	2024–2025
Independent Management Consultant	2024–present

Research Question

How can you improve investments into Start-Ups and SMEs through legislative change of Investment Modalities?

- ▶ AI at the forefront of Innovation
- ▶ *“Automating Cognitive Workflows”*
- ▶ Recipient of EXIST Grant

Do the signals used to select firms generate the value public funding is meant to create?

- ▶ Which observable signals enter the funding decision?
- ▶ Do these signals predict innovation, spillovers, and economic value?
- ▶ Do programmes weight signals differently when they define value differently?

The project separates the **selection mechanism** from the grant itself.

Theoretical framework: selection signals as policy instruments

The project treats selection signals as **proxies for expected public value**.

Policy objective → Selection signals → Funding decision → Outcome layers

- ▶ Programmes define value differently: innovation, spillovers, or local economic value.
- ▶ Granting authorities translate these objectives into signal weights.
- ▶ The empirical question is whether these weights predict the outcomes they are meant to generate.

The grant is therefore not the only treatment; the selection rule is part of the mechanism.

Empirical framework



The unit of analysis is the **firm–application**. The project links three layers of data:

- ▶ **Selection signals:** founder, project, technology, and regional characteristics
- ▶ **Selection rule:** how programmes weight these signals in the funding decision
- ▶ **Outcome layers:** innovation, spillover potential, and economic value

The empirical object is not the grant alone, but the link between **what programmes select for** and **what selected firms later generate**.

Identification strategy

Identification comes from variation in **selection rules**, not only from grant receipt. The project compares how different programmes weight observable signals and whether those weights predict later outcomes.

- ▶ Compare applicants within programmes and calls
- ▶ Use rankings, thresholds, or score cutoffs where available
- ▶ Compare programmes that define public value differently
- ▶ Test alignment between signal weights and realised outcome layers

Where clean cutoffs exist, the design can estimate local causal effects. Otherwise, the core estimate is the **selection–outcome alignment gap**.

Identification strategy

Identification comes from **variation in selection rules**.

- ▶ within-call applicant comparisons
- ▶ rankings, thresholds, score cutoffs
- ▶ programmes with different value definitions
- ▶ alignment between signal weights and outcomes

Fallback estimand: **selection–outcome alignment gap**

Expected Contribution

- ▶ **Empirical:** estimate whether selection signals generate the outcomes used to justify public support.
- ▶ **Conceptual:** separate the effect of receiving a grant from the design of the rules that allocate grants.
- ▶ **Policy-relevant:** identify where programme objectives and actual selection practice diverge.
- ▶ **PhD logic:** the project can begin as one empirical paper on grant selection design and develop into a broader research agenda on innovation policy.

Backup: Data Sources

- ▶ Programme documents, eligibility criteria, scoring rubrics and beneficiary lists.
- ▶ Applicant-level award data from CORDA / ORBIS where accessible.
- ▶ Patent data from EPO / PATSTAT.
- ▶ Firm accounts, business registers and regional employment data.
- ▶ Score-level data from programmes such as the EIC Accelerator where available.

Backup: Identification Risks

- ▶ Self-selection into application.
- ▶ Score manipulation near cut-offs.
- ▶ Missing or incomplete score-level data.
- ▶ Survivorship bias in firm-level outcomes.
- ▶ Outcome correlation across innovation, spillover and economic-value layers.